Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	25 September 2026
Team ID	23KD1A1507
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	Authentication via ServiceNow SSO and Service Portal login (`/sp`).
		Role-based access mapping for Requester, Manager, Security, and Fulfiller roles.
FR-2	Service Catalog Intake	Network Request catalog item created under Maintain Items in the Network category.
FR-3	Form Design & Variables	Dynamic form UI policies to show/hide connection type and access level fields.
		Variable sets for Requester Info (Requested For, Mobile Number, Customer Address).
FR-4	Database Table & Schema	Custom Database table creation (`u_database_table`) for request records.
		Auto-numbering field configuration with prefix and formatted sequence (`Database Number`).
FR-5	Flow Designer Automation	Trigger on Service Catalog submission and map variables via Get Catalog Variables.
		Automated record creation in custom database table (`u_database_table`).
FR-6	Approval Workflow	Multi-level approval routing using Ask for Approval (Manager & Network Security).
		Flow logic branch handling for Approved and Rejected request states.
FR-7	Email Notifications	Trigger Send Email action upon record creation and approval status change.
		System Email Logs verification (Sent/Received logs tracking).
FR-8	Task Fulfillment	Automatic creation and assignment of fulfillment tasks (`u_network_task`).
		Update Record action in Flow Designer to synchronize ticket closure state.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-9	Access Control & Security	Configured Access Control Rules (ACLs) to restrict read/write access to sensitive fields.
		Role-based permissions for Network Database and Task tables.
FR-10	Data Integrity & Validation	Client-side and server-side mandatory field enforcement to prevent incomplete forms.
		Auto-populating requester location and user information using Catalog Client Scripts.
FR-11	Reporting & Troubleshooting	Flow Execution Details tracing and execution path analysis.
		Operations reporting dashboard for request volume, fulfillments, and SLAs.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Intuitive Service Portal interface (`/sp`) with dynamic forms, clear error messaging, auto-populated requester details, and minimal click-depth to enable non-technical users to submit requests effortlessly.
NFR-2	Security	Role-Based Access Control (RBAC) enforced through ServiceNow ACLs; HTTPS encryption for all data in transit; field-level permissions preventing unauthorized modification of network database records (`u_database_table`).
NFR-3	Reliability	Transaction processing with full data integrity; mandatory field checks and approval state conditions prevent incomplete requests; full audit history logging for all record creations and workflow state transitions.
NFR-4	Performance	Flow Designer triggers workflow actions within 3 seconds of catalog submission; automated email notifications generated within 5 seconds; real-time status updates rendered smoothly on the Service Portal.
NFR-5	Availability	24x7 cloud availability hosted on the ServiceNow Personal Developer Instance (PDI) with target uptime of 99.5%; persistent record logging ensures submitted tickets are retained safely across sessions.
NFR-6	Scalability & Maintainability	Built using native ServiceNow Flow Designer and Creator Studio low-code features without custom script includes; allows simple addition of new network categories, teams, or approval chains without core code modification.